

MGTF 413: Computational Finance Methods

TERM (Winter 2026)

This course will be co-taught by the following two instructors.

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Class Meeting Time: Fridays, 1-3:50pm (PST)

DESCRIPTION

Computational methods have become indispensable in modern finance. This course will introduce common computational methods of importance for finance and illustrate their use in solving problems. The course will begin with an introduction to optimization, which will include application of quadratic programming to portfolio selection. This will be followed by an introduction to simulation methods with applications to option pricing. Fundamentals of Monte Carlo simulation and numerical solution of stochastic differential equations will be treated. The course will conclude with a discussion of finite difference methods for solving partial differential equations arising in finance. Applications to pricing of European, American and exotic options will be given using simulation and partial differential equation methods.

OBJECTIVES

At the conclusion of this course, the student will:

- understand the role of optimization problems and techniques in finance;
- understand the distinction between unconstrained, constrained, nonlinear, quadratic programming (QP)
- understand how to formulate and numerically solve QP problems;
- be able to apply basic algorithmic tools for QP to portfolio optimization and other problems in finance;
- understand the basic ideas of Monte Carlo techniques for the purpose of option pricing;
- understand basic concepts related to simulation including random variable generation, variance reduction methods and statistical analysis of simulation output;
- understand and be able to apply the computational methods of importance sampling, pricing using binomial trees, explicit and approximate solutions of stochastic differential equations including Euler and Milstein schemes;
- understand the basic concepts of numerical quadrature;
- be able to use Monte Carlo simulation in solving applied problems on derivative pricing, including but not limited to pricing of European and American options, pricing interest rate dependent claims and path dependent options;
- understand the role of partial differential equations in option pricing;
- understand how to formulate and numerically solve the Black-Scholes equation for European and American options;
- be able to use basic numerical methods for solving the Black-Scholes partial differential equation for option pricing;
- understand how the Black-Scholes framework can be extended to exotic options and nonlinear models.

MATERIALS

-Course notes, which are required reading, will be provided in electronic format via the course website on Canvas at <https://rady.instructure.com/>

-Students are expected to read the notes and other relevant material provided on Canvas in advance of the class meeting time.

The course will use Python and R within Jupyter notebooks.

Reference Materials: for extended reading by interested students (electronic access available through roger.ucsd.edu --- use vpn to gain access requiring a UCSD IP address)

1. [Optimization Methods in Finance](#), G. Cornuejols and R. Tutuncu, Second Edition, Cambridge Univ. Press, 2018.
2. [Introduction to the Mathematics of Finance](#), R J Williams, American Mathematical Society, 2006.
3. [Implementing Models in Quantitative Finance: Methods and Cases](#), Gianluca Fusai, Andrea Roncoroni, Springer, 2008
4. [Monte Carlo Methods in Financial Engineering](#), Paul Glasserman, Springer, 2003.
5. [Tools for Computational Finance, Sixth Edition](#), R. U. Seydel, Springer, 2017

COURSE SCHEDULE

Week 1

- examples of optimization problems and techniques in finance

- linear programming, with applications to asset/liability cash flow matching, arbitrage detection, and bond portfolio management
- sensitivity analysis

Week 2

- Newton's method
- unconstrained optimization, steepest descent, Newton's method, interior-point method, and sequential quadratic programming
- quadratic programming (QP): portfolio mean-variance optimization

Friday of Week 2: Homework Assignment 1a due

Week 3:

- discrete models of option pricing
- dynamic programming
- pricing and hedging with binomial trees: European call and American put

Friday of Week 3: Homework Assignment 1b due

Week 4

- motivation for Monte Carlo techniques in computational finance
- static Monte Carlo
- simulating random variables

Friday of Week 4: Homework Assignment 2a due

Week 5

- variance reduction
- importance sampling
- applications to option pricing in discrete time models

Friday of Week 5: Homework Assignment 2b due

Week 6

- dynamic Monte Carlo
- simulating continuous stochastic processes: Euler and Milstein schemes
- pricing European options
- numerical quadrature and pricing Asian options

Friday of Week 6: Homework Assignment 3a due

Week 7

- pricing American options
- dynamic programming and Longstaff-Schwartz algorithm

Friday of Week 7: Homework Assignment 3b due

Week 8

- **COMPREHENSIVE EXAM** (1.5 hours)
- basic notation and concepts for working with partial differential equations
- examples of partial differential equations
- the Black-Scholes partial differential equation for European options

Friday of Week 8: Homework Assignment 3c due

Week 9

- finite difference methods for numerical approximation: explicit, implicit, and Crank-Nicolson schemes
- stability, consistency, convergence, and accuracy of numerical methods

Week 10

- the Black-Scholes free boundary problem for American options
- numerical methods for American option pricing under the Black-Scholes model
- extensions to exotic options and nonlinear models

Friday of Week 10: Homework Assignment 4 due

Week 11

Final project due: paper and presentations (about 15 min in length) to be given in final exam time slot of Friday, March 20, 2026, 1-4pm

COURSE GRADES

Course grades will be based on the following weighted components:

Weekly quizzes (on Canvas): 10%
Written homework assignments: 4 in total (28%)
Comprehensive (midterm) exam in week 8: (30%)
Final project: written report and presentation (32%)

Homework Policy

Students are strongly encouraged to first attempt the homework assignments on their own. After that, students may consult other students, the Professors, or the TA.

However, **students must produce their own individual final solutions independently.**

Students must NOT copy solutions from another student or from any other source.

Do NOT share your own independent homework solution with anyone else.

If a student got an idea for the homework from another student, the name of that student and the extent of the helpful idea must be acknowledged on the first page of the homework solution.

Moreover, all communication with the TA and Professors needs to be done through UCSD e-mail/Piazza accounts only.

Final Project

The final project should be completed in groups of 5-6 people. The contribution of each member of the group needs to be identified in the written report and each member of the group needs to participate in the final presentation.

CLASS POLICIES

The highest professional standards are expected of all members of the Rady community. The collective class reputation and the value of the Rady experience hinges on this.

Student learning and the classroom experience is enhanced when:

- Students join class on time. On time arrival ensures that classes are able to start and finish at the scheduled time. On time arrival shows respect for both fellow students and faculty and it

enhances learning by reducing avoidable distractions. This will enable the instructors to have more interaction with the class and will provide a good experience for the whole class.

- Students are fully prepared for each class meeting. Students will gain much more from the class meetings and associated discussions if they read the notes in advance of the class meeting time. When students are not prepared they cannot contribute to the overall learning process. This affects not only the individual, but their peers who count on them, as well.
- Students respect the views and opinions of their colleagues. Discussion is encouraged. Intolerance for the views of others is unacceptable.
- Students minimize distractions while participating in class meetings. When students are surfing the web, responding to e-mail, instant messaging each other, and otherwise not devoting their full attention to the topic at hand, they are doing themselves and their peers a major disservice. Fellow students cannot benefit from the insights of students who are not engaged. Phones and other devices not being used for the class session should be put away. Their use is not professional and takes attention away from class participation.

Class materials and recordings: Class materials must not be shared beyond the class. In particular, notes and recordings (audio, photo or video) made by the instructors are not to be shared. No recordings (audio, photo or video) should be made by students. Students are encouraged to take notes.

ACADEMIC INTEGRITY

Integrity of scholarship is essential for any academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of University intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind.

Further elaboration is provided in the following statement modeled after one provided by the [UCSD Academic Integrity Office](#):

"Academic Integrity is expected of everyone at UC San Diego. This means that you must be honest, fair, responsible, respectful, and trustworthy in all of your actions. Lying, cheating or any other forms of dishonesty will not be tolerated because they undermine learning and the University's ability to certify students' knowledge and abilities. Thus, any attempt to get, or help another get, a grade by cheating, lying or dishonesty will be reported to the Academic Integrity Office and will result in sanctions. Sanctions can include an F in this class and suspension or dismissal from the University. So, think carefully before you act by asking yourself: a) is what I'm about to do or submit for credit an honest, fair, respectful, responsible & trustworthy representation of my knowledge and abilities at this time and, b) would my instructor approve of my action? You are ultimately the only person responsible for your behavior. So, if you are unsure, don't ask a friend—ask your instructor, instructional assistant, or the Academic Integrity Office. You can learn more about academic integrity at academicintegrity.ucsd.edu"

For tips on how to Excel with Integrity, check out the [Academic Integrity webpage here](#).

The complete UCSD Policy on Integrity of Scholarship can be viewed at:
<http://senate.ucsd.edu/Operating-Procedures/Senate-Manual/Appendices/2>

ASSESSMENT VERSIONING

Following UCSD (and common) practice recommended by the Academic Integrity Office, assessments, especially those given at non-overlapping times, will be comparable, but may not be identical. This practice is meant to maintain course integrity, avoiding non-allowed collaboration (either intentional or accidental).

STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the class must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member.

The student must provide the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member as early as possible in order to initiate the request for accommodation in classes, examinations, or other academic program activities. **No accommodations can be implemented retroactively.**

Please visit the [OSD website](#) for further information or contact the Office for Students with Disabilities at (858) 534-4382 or osd@ucsd.edu.